Free fall

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe and calculate ideal free fall near Earth.

SUCCESS CRITERIA - I can...

1. State that you are ignoring air resistance.
2. Use equal velocity changes in equal times.
3. Separate acceleration from mass and current velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **4 kg** cart starts from rest under **24 N** net force right for **3 s**.
1. **Find acceleration** and final velocity.
2. **State one thing you** are treating as true.

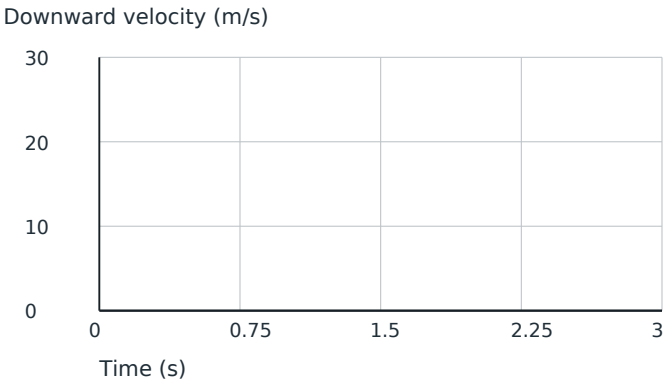
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $v_f = v_i + g\,\Delta t$.
Downward positive; g = **10 m/s²**; gravity only.

Worked reasoning: A ball dropped from rest has downward speeds **0**, **10**, **20 m/s** at **0**, **1**, **2 s** in ideal free fall with g=**10 m/s²**. Equal times give equal velocity changes.

Your turn: 1. **Sketch its velocity-time graph** with downward positive.
2. **Explain why the positions** are not equally spaced.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two balls of different mass are released together in a vacuum from the same height. **Compare acceleration** and arrival time.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball is released from rest in a drop simulation. Ignore air resistance and use g = **10 m/s²**. **Complete the table**, then **describe how the velocity changes** each second.

Time (s)	Downward velocity (m/s)
0	0
1	____
2	____
3	____

Marzano check - circle: ① ② ③

05 Explain your reasoning

At the highest point of a tossed ball, a classmate says gravity stops because speed is zero. Correct the statement.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A fixed-mass cart has acceleration **4 m/s²** under **8 N** net force. **Find acceleration** when net force becomes **12 N**. Explain.
Use the table to keep the controlled and changed quantities distinct.

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

07 Exit check

A dropped ball falls from rest for **4 s** in an air-free simulation.
1. Use g = **10 m/s²**.
2. **Find its final velocity** and **add a correctly directed arrow** to the diagram.

Dropped ball

State your positive direction. →

Velocity:

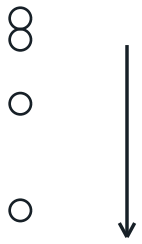
Acceleration / force:

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Draw three equal-time snapshots of a dropped ball. Then **draw a heavier ball released** with the same setup in a vacuum. **Compare the patterns**.

Equal time intervals



Marzano check - circle: ① ② ③